

autoPET V — rogerking

Cross-Round Aggregation in Interactive Whole-Body PET/CT Lesion Segmentation: a Training-Free Study Alongside a Baseline Submission to autoPET V

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autoPET V challenge (MICCAI 2026) — team rogerking.

Algorithm as submitted on Grand Challenge: “nnU Net interactive baseline (autoPET V)”. This report follows the structure of the official autoPET_Template.pdf.

Abstract

autoPET V scores interactive whole-body PET/CT lesion segmentation by the area under the Dice-versus-interaction-round curve, so the trajectory across rounds matters as much as the final state. We submit the organisers’ nnU-Net interactive baseline without modification, and report alongside it an offline study of training-free strategies that reuse predictions from earlier interaction rounds. On the single worked example released with the challenge repository, a running union of predictions across rounds raises the AUC of Dice from 3.5141 to 3.5484 and converts a non-monotone Dice trajectory into a monotone one, which is precisely what an area-under-curve metric rewards. **Submitted to the challenge’s preliminary test phase, however, that variant ranked eight places below the unmodified baseline (56th against 48th), so it was not carried forward** — a single-case effect with a plausible mechanism that did not survive held-out data. Every tightening or smoothing aggregate we tested was worse than the union, and one apparently

reasonable variant — deleting whole connected components touched by a background scribble — was catastrophic (AUC 2.7795), because under accumulation the lesion region merges into a single large component. Decomposing the union’s residual error shows that 78.6 % of it consists of voxels the model never predicted in any round, which bounds what any inference-time aggregation can achieve and is our reason for not carrying this into the submission. All numbers come from one case, one scribble strategy and one tracer; they are reported as a hypothesis-generating observation, not as a validated result.

Keywords: autoPET challenge · interactive segmentation · whole-body PET/CT · nnU-Net · area under the Dice curve

1 Introduction

autoPET V addresses interactive segmentation of tumour lesions in whole-body PET/CT. An automated annotator inspects the current prediction, locates the largest error region, and simulates a scribble there: a foreground scribble when the dominant error is under-segmentation, a background scribble when it is over-segmentation. Scribbles accumulate, so round t sees every scribble from rounds 1 to $t-1$, and three scribble strategies (centerline, random, boundary) are used in equal proportion across the test set.

The score is the trapezoidal area under the Dice-versus-round curve rather than the Dice of the final round. Two consequences follow, and they shaped what we examined. First, a dip in any intermediate round is subtracted from the area directly, so a method that oscillates is penalised relative to one that improves monotonically to the same endpoint. Second, under the trapezoidal rule the first and last rounds carry weight 0.5 while intermediate rounds carry weight 1.0, so the early rounds are worth more than the last — the opposite of the usual intuition that interactive segmentation is about where one finishes.

2 Methods

The submitted algorithm is the organisers' nnU-Net interactive baseline, unmodified. We state this at the outset because the study in Section 3 was carried out offline on stored predictions and forms no part of the submitted container.

2.1 Data

Training data. None were used by us: no model was trained for this submission. The submitted weights are the organisers' released Data set998_AutoPETV checkpoint, trained by them on 1611 cases drawn from the autoPET series of whole-body PET/CT datasets with manually annotated lesions [1], covering FDG PET/CT, PSMA PET/CT and related collections.

Validation data. The offline study in Section 3 uses the single worked example shipped in the challenge repository (a PSMA case) together with the organisers' own stored predictions and Dice values for five interaction rounds. No additional data were downloaded or annotated.

2.2 Data pre-processing

nnU-Net's own default preprocessing, unchanged. The container converts the challenge's `lesion-clicks.json` into foreground and background scribble volumes following the SW-FastEdit convention for interactive whole-body PET annotation [2], and writes four input channels: CT, PET, foreground-scribble heat map, background-scribble heat map.

2.3 Algorithm/model

nnU-Net v2 [3], configuration `3d_fullres`, `nnUNetTrainer__nnUNetPlans__3d_fullres`, two output classes (background, tumour). Inference is `nnUNetv2_predict -d 998 -c 3d_fullres -f 0 --disable_tta`; the lesion mask is written to the challenge's expected output interface.

2.4 Data post-processing

None. The container emits nnU-Net's output unchanged. The post-processing strategies studied in Section 3 were deliberately **not** included; the reasoning is given in Section 4.

2.5 Training and test parameters

No training was performed, so no learning rate, schedule or augmentation policy is ours to report; the organisers' training configuration applies. At test time, fold 0 is used alone, test-time augmentation is disabled, and no ensembling of any kind is performed.

Table 1. Algorithm details.

field	value
Team name	rogerking
Algorithm name (as submitted on Grand Challenge)	nnU Net interactive baseline (autoPET V)
Data pre-processing	nnU-Net default normalisation; scribbles rendered to two additional input channels
Data post-processing	none
Training data augmentation	not applicable (no training performed)
Test time augmentation	none (--disable_tta)
Ensembling	none (single fold, fold 0)
Standardised framework	nnU-Net v2
Network architecture	U-Net (3D), 4 input channels
Loss	DSC + CE (organisers' training)
Training data	organisers' Dataset998_AutoPETV, 1611 cases
Data/model dimensionality	3D
Use of pre-trained models	yes — the organisers' publicly released fold-0 checkpoint
GPU hardware for training	none used by us

2.6 Github repository

<https://github.com/Rogerking928/autopet-v-rogerking> (Apache-2.0)

The repository contains everything needed to rebuild the submitted container — Dockerfile, build and export scripts, process.py, utils.py — together with the model weights as a release asset (weights.zip, the organisers' released Dataset998_AutoPETV fold-0 checkpoint, redistributed under Apache-2.0 with attribution). It also contains the analysis scripts that reproduce every number in Section 3 (analysis/temporal.py, analysis/error_budget.py, analysis/scribble_postproc.py, analysis/cache_roundtrip.py), which operate on the predictions already stored in the challenge repository and require neither a GPU nor the model weights.

The analysis was written and run by the author in Python 3, with Claude (Anthropic) and ChatGPT (OpenAI) used as assistants under the author’s direction; the author reviewed and tested all code and takes full responsibility for the results it produces.

3 Results

No cross-validation results can be reported: no model was trained, so there is no training split to cross-validate. What follows are the preliminary-phase results of the submitted system and of one variant, and then the offline study that motivated that variant.

3.0 Preliminary leaderboard

Two containers were submitted to the preliminary test phase:

container	preliminary rank	mean position
organisers’ baseline, unmodified (submitted)	48 / 157	54
baseline + running union across rounds (not submitted)	56 / 158	57.5

The running union, which improved the area under the Dice curve on the single offline example, made the leaderboard rank worse by eight places. This result is the reason the submitted container carries no post-processing at all, and it is discussed in Section 4.2.

The comparison also settles a question we could not answer offline. The two containers differ only in whether the current prediction is unioned with the previous rounds’ predictions carried through the container’s /cache/ volume. If that volume were not persisted between interaction rounds, the union would never fire and the two submissions would be numerically identical. They are not, so the cache does persist on the evaluation platform — and the union, having fired, was harmful.

3.1 Setup, and the limits of this setup

Every number below is a post-hoc recombination of five stored binary masks from a single case. No model was run. Three limitations disqualify any strong claim, and we state them before the results rather than after:

1. **n = 1 case.** Differences below roughly 0.01 AUC cannot be separated from case-level idiosyncrasy.
2. **One scribble strategy.** The stored example uses random; the test set mixes centerline, random and boundary in equal thirds.
3. **One tracer.** The example is PSMA; no FDG case was available to us.

We report the study because the direction of the effects is consistent and because the negative results are instructive, not because the magnitudes are trustworthy.

3.2 Cross-round aggregation

Table 2. Training-free aggregation strategies, AUC of Dice over five rounds.

strategy	AUC of Dice	Δ vs baseline
baseline (organisers' stored predictions)	3.5141	—
running union of rounds 1...t	3.5484	+0.0343
union of last 4 rounds	3.5478	+0.0338
union of last 3 rounds	3.5445	+0.0304
union + local background-scribble pruning ($r = 2$)	3.5425	+0.0284
keep-best-prefix (post-hoc oracle)	3.5360	+0.0219
union of last 2 rounds	3.5325	+0.0184
EMA, $\alpha = 0.5$	3.5258	+0.0117
majority vote	3.5242	+0.0101
EMA, $\alpha = 0.7$	3.5141	± 0.0000
scribble-consistency post-processing (best of 10 settings)	3.5141	± 0.0000
EMA, $\alpha = 0.3$	3.4991	-0.0149
running intersection	3.4496	-0.0644
union + component-level background pruning	2.7795	-0.7346

The running union is the only strategy that clearly helps, and it helps in the way the metric rewards: the Dice trajectory becomes monotone (0.870, 0.887, 0.888, 0.891, 0.894), removing the baseline's dip at round 2. It also exceeds keep-best-prefix, a post-hoc oracle that selects the best prefix length using the ground truth — possible only because the union recovers true positives that no single round contains on its own.

The mechanism follows from scribble accumulation: because scribbles accumulate but the model re-predicts from scratch each round rather than editing its previous output, different rounds recover different true positives, and the union merges them. The risk is equally clear — a union only grows, so a false positive entering at any round is locked in for the remainder.

3.3 Why the union saturates

Decomposing the residual error of the full union into voxels the model never predicted in any round (FN_never) and voxels the union added incorrectly (FP_union):

- **FN_never = 3,622 voxels = 15.7 % of the ground truth = 78.6 % of the total error**
- FP_union = 988 voxels = 4.3 % of the ground truth = 21.4 % of the total error

More than three quarters of the residual error consists of voxels no round ever touched, so no inference-time recombination of those rounds can recover them.

3.4 Negative results

Component-level deletion is a single point of failure. Deleting the entire connected component touched by a background scribble — the natural reading of “the annotator says this object is background” — cost 0.73 AUC. Under a running union the lesion region merges into one very large connected component, so a single background scribble on its edge deletes the whole prediction: round-2 Dice fell from 0.886 to 0.135, with exactly one component removed.

Scribble-consistency post-processing does nothing. Forcing the prediction to agree with the scribbles was our original idea; all ten parameter settings were neutral or harmful and the optimum

was radius 0, i.e. doing nothing. Two observations explain this. Background scribbles removed zero components in all five rounds, because the non-accumulating baseline had already stopped predicting at those locations by the following round. And the more voxels a foreground scribble added, the worse the result ($128 \rightarrow 776$ voxels, $\Delta -0.0024 \rightarrow -0.0081$), because the simulated scribbles run along the lesion centreline where the model is already correct, so padding around them adds only false positives. The model's errors are at the boundary; the scribbles are not.

Tightening and smoothing move the wrong way. A running intersection lost 0.0644 with monotonically falling Dice, confirming that the error structure is dominated by under-detection rather than over-detection. Exponential moving averages were non-monotone in α ($-0.0149, +0.0117, \pm 0.0000$ for $\alpha = 0.3, 0.5, 0.7$), which we read as noise rather than signal.

4 Discussion

4.1 Independent support for the scribble-encoding hypothesis

Section 3.4 concluded that the model's errors are at lesion boundaries while the simulated scribbles run along lesion centrelines, and Section 4.2 below argues that the baseline's $\sigma = 0$ scribble heat maps — which reduce each scribble to isolated voxels in the input channels — are a plausible binding constraint.

The public preliminary leaderboard offers indirect support. Six of the top eight entries at the time of writing are submitted under the name “AutoPET V interactive lesion segmentation (ResEncL + EDT clicks)”, i.e. a residual-encoder nnU-Net configuration combined with a Euclidean-distance-transform encoding of the clicks rather than isolated-voxel heat maps. That the leading entries changed exactly the component our error analysis identified is consistent with, though not proof of, the interpretation above. It also locates the difference in training rather than in inference: a distance-transform encoding is out of distribution for a network trained on $\sigma = 0$ maps, so adopting it requires retraining.

4.2 What we did and did not carry into the submission

If these observations hold beyond one case, the practical implication is narrow but concrete: because the metric integrates over rounds and different rounds recover different true positives, carrying a running union in the container's persistent cache is a training-free change that improves the early rounds, which the trapezoidal rule weights most heavily.

We did not include it in this submission, and the reason is no longer a judgement call: we built it, submitted it to the preliminary phase, and it was worse (Section 3.0, rank 56 against 48). The most likely explanation is the risk we had already identified — a union only grows, so any false positive it admits is locked in for the remaining rounds. On one case the true positives it recovered outweighed that cost; across the preliminary test set they did not.

This is the clearest lesson we can offer from this submission, and it is a lesson about method rather than about PET: an effect of +0.0343 AUC measured on a single case, with a mechanism that seemed sound and a metric that seemed to favour it, did not survive contact with a held-out test set. We reported the offline magnitudes in Section 3.2 with three explicit disclaimers; the disclaimers were the correct reading and the effect was not.

The error decomposition points elsewhere for real improvement. With 78.6 % of the union's residual error consisting of never-detected voxels, the binding constraint is detection sensitivity, not the reconciliation of predictions across rounds. One candidate we could not test is the scribble representation itself: the baseline generates its scribble heat maps with `sigma = 0`, reducing each scribble to isolated voxels in the input channels. Giving scribbles a non-zero spatial extent would give the network something it can actually see, but this requires retraining and was out of scope here.

5 Conclusion

We submit the organisers' interactive nnU-Net baseline unmodified. An offline study on the single released example suggested that a running union across interaction rounds was the only training-free aggregation that improved the area under the Dice curve, and that it

did so by removing an intermediate dip rather than by raising the endpoint. Submitted to the preliminary phase, that variant ranked eight places below the plain baseline, so it was not carried forward.

The transferable content of this report is therefore negative throughout: a single-case effect of +0.0343 AUC that reversed on held-out data; the catastrophic failure of component-level deletion under prediction accumulation (−0.73 AUC); the complete ineffectiveness of scribble-consistency post-processing, whose optimum over ten settings was to do nothing; and an error decomposition showing that 78.6 % of the residual error consists of voxels the model never predicts in any round, which bounds what any inference-time method can achieve here.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) and ChatGPT (OpenAI) to draft and copy-edit prose. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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